

SYGIC TELEMATICS — FULL ARCHITECTURE

.NET 8 | Docker | Kubernetes | gRPC | GraphQL | YARP | RabbitMQ | MassTransit

VMWARE WORKSTATION 17 PRO — ON-PREMISES INFRASTRUCTURE

k8s-master

Ubuntu Server 24.04 LTS
192.168.1.100

Control Plane + Registry + Docker

k8s-worker1

Ubuntu Server 24.04 LTS
192.168.1.101

Worker Node

k8s-worker2

Ubuntu Server 24.04 LTS
192.168.1.102

Worker Node

GraphQL (HotChocolate)

REST / HTTP

gRPC (HTTP/2 + Protobuf)

YARP Reverse Proxy

RabbitMQ Events

Microservice

SQL Server Database

Docker Container

Kubernetes / VMware

KUBERNETES CLUSTER (KUBEADM + FLANNEL CNI)

Flannel Pod Network — 10.244.0.0/16

NAMESPACE: SYGIC-TELEMATICS

External Client

Postman / Browser / Angular App

sygic.local

NGINX Ingress Controller

sygic.local → gateway-api:8080 | NodePort 30080/30443

API GATEWAY (gateway-api:8080)

GraphQL

/graphql (HotChocolate)

BFF Controllers

REST /api/bff/*

YARP Proxy

Reverse Proxy

Auth (JWT)

Bearer Token

HTTP / REST

gRPC (HTTP/2)

RabbitMQ Events

Identity

:8080

HTTP

Vehicle

:5103 / :6103

HTTP

gRPC

Location

:5104 / :6104

HTTP

gRPC

Battery

:5105 / :6105

HTTP

gRPC

Trip
:5106 / :6106
HTTP gRPC

Telemetry
:5107 / :6107
HTTP gRPC

Alert
:5108 / :6108
HTTP gRPC

IdentityDB

VehicleDB

LocationDB

BatteryDB

TripDB

TelemetryDB

AlertDB

SQL Server 2022
Deployment + PVC (10Gi) + ClusterIP Service
SA password in K8s Secret

RabbitMQ 3 + Management
Deployment + PVC (2Gi) + ClusterIP Service
MassTransit Async Event Bus

PersistentVolumeClaim
sqlserver-pvc — 10Gi

PersistentVolumeClaim
rabbitmq-pvc — 2Gi

ConfigMap
RabbitMQ host, ports, settings

Secrets
DB connection strings, JWT TokenKey, SA password

Private Docker Registry
192.168.1.100:5000 | registry:2 | HTTP (insecure)

DOCKER COMPOSE — LOCAL DEVELOPMENT (DOCKER-COMPOSE.YML)

gateway-api
:5100

identity-api
:5101

catalog-api
:5102

vehicle-api
:5103/:6103

location-api
:5104/:6104

battery-api
:5105/:6105

trip-api
:5106/:6106

telemetry-api
:5107/:6107

alert-api
:5108/:6108

sqlserver
:1434→1433

rabbitmq
:5672/:15672

Network: **sygic-net** (bridge) | Volumes: **sqlserver-data**, **rabbitmq-data** | Environment: **ASPNETCORE_ENVIRONMENT=Docker**

Build & Deploy Pipeline

1. dotnet publish

→ 2. docker build

→ 3. docker push
192.168.1.100:5000

→ 4. kubectl apply
k8s manifests

